Dr Aakash Bansal

Lecturer in Applied Electromagnetics, Loughborough University

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Research Profile

Early-career academic in applied electromagnetics working at the intersection of reconfigurable antennas, metamaterials, secure and resilient networks for (non-)terrestrial communications. Research programme spans active millimeter-wave beam-steering antennas, 3D-printed RF frontend systems, reconfigurable intelligent surfaces, secure networks, orbital angular momentum waves, from electromagnetic theory through to fabricated, measured hardware. Leading a group of eight researchers (3 PDRAs, 4 PhDs and 2 PG) and has secured over £4M in research funding as PL/Co-L since appointment in 2024. Authored 80+ peer-reviewed publication with several close to commercialisation.

Academic Appointment and Work Experience

Deputy Director of Doctoral Programmes Jul 2026 – Present
Wolfson School of Mechanical, Electrical, and Manufacturing Engineering, Loughborough University, UK
- *School-level responsibility for doctoral recruitment, student wellbeing and progression, resolution of supervisory conflicts, and support colleagues in securing doctoral funding. Oversight of approx. 150 doctoral researchers.*

Lecturer in Applied Electromagnetics (equiv. Assistant Professor) Sep 2024 – Present
Wolfson School of Mechanical, Electrical, and Manufacturing Engineering, Loughborough University, UK
- *Independent research programme in reconfigurable antennas, metasurfaces, networks, and satellite communications; doctoral and post-doctoral supervision; research funding acquisition; co-management of RF facilities at the university.*
- *Module delivery and development in microwave engineering, communication engineering, and applied electromagnetics.*

Royal Academy of Engineering UK IC Research Fellow Jan 2024 – Dec 2025
Wolfson School of Mechanical, Electrical, and Manufacturing Engineering, Loughborough University, UK
- *Two-year independent fellowship under the RAEng scheme for UK Intelligence Community; aimed at developing low-power beam-steering antennas for CubeSat platforms with multiple industrial partners.*

Research Associate in RF Systems Feb 2023 – Dec 2023
Wolfson School of Mechanical, Electrical, and Manufacturing Engineering, Loughborough University, UK
- *Named RA for UKRI, DSTL, and industry funded projects to develop orbital angular momentum wave generation metasurface, optically transparent antennas, and high-resolution imaging.*

Future Communications Engineer Oct 2022 – Jan 2023
Satellite Applications Catapult, UK
- *Secondment directly under CTO; developed a Ku-band beam-steering antenna for OneWeb satellite handover at ground stations.*

Research Assistant Apr 2019 – Sep 2022
Wolfson School of Mechanical, Electrical, and Manufacturing Engineering, Loughborough University, UK
- *Research support for the group; contribution to UKRI and industry-funded projects; design, fabrication and measurement of different RF front-end systems including antennas and metasurfaces.*

Research Associate Jul 2017 – Feb 2019
CSIR – Central Electronics Engineering Research Institute, India
- *Dynamic modelling of staggered double-vane slow-wave structures and beam-wave interaction analysis for planar travelling wave tube amplifiers.*

Education

PG Certificate in Learning and Teaching in Higher Education, Loughborough University, UK 2025 – 2027
Postgraduate certificate course for new academics, leading to Fellowship of Advanced HE; expected completion in Mar 2027.

PhD in Electrical, Electronics and Systems Engineering, Loughborough University, UK 2019 – 2022
- *Dissertation on Active Beam-Steering millimeter-wave Antenna System for 5G and Beyond (DOI).*

Bachelor of Technology, Guru Gobind Singh Indraprastha University, India 2013 – 2017
- *Majored in Electronics and Communications Engineering with 78.3% and ranked among top 15 out of 200 students.*

Research Funding

Received > £4M in research funding since 2024 with > £1.5M as Project Lead and > £2M directly to host institution.

Responsible and Resilient Electronic and Electromagnetic (RaRE) Materials Network+ Oct 2026 – Mar 2030
Co-lead with value of £1.38M from UKRI Engineering and Physical Sciences Research Council
- *UK-wide network on materials for resilient electronics and electromagnetic systems;*

RCS Reduction using Metasurfaces for Unmanned Aerial Vehicles Sep 2026 – Feb 2027
Lead with value of £270k from UK Defence and Innovation with Lockheed Martin UK
- *Thin film metasurfaces to reduce radar cross-section and increase UAS survivability*

Neuron-Inspired Super Smart Textiles for Remote Health Monitoring Jul 2026 – Jun 2028
Co-lead with value of £510k from UKRI Engineering and Physical Sciences Research Council
- *Textile-based sensors embedded in hospital clothing and athlete outfits for remote health monitoring*

C-Band Intelligent Antenna Systems for PNT in Remote Areas Oct 2025 – Mar 2027
Lead with value of £605k from European Space Agency with GMV UK
- *Intelligent C-band beamforming antenna sustaining gain in adversarial scenarios for PNT – aimed for commercialisation.*

Diverse Research in Inclusive Vehicle Environments Mini CDT Oct 2025 – Sep 2028
Co-lead with estimated value of £750k from Loughborough University, UK
- *Five internally funded PhD studentships for interdisciplinary research on inclusive autonomous vehicles.*

UK-US Alliance on New Physical Layer Security for Wireless Infrastructure Oct 2015 – Mar 2026
Lead with value of £100k from UKRI Engineering and Physical Sciences Research Council
- *Threat and vulnerability identification for 6G infrastructure, focused on millimeter-wave reconfigurable intelligent surfaces.*

Additively Manufactured RF Frontend Systems – PhD Studentship Oct 2025 – Sep 2028
Co-Lead with value of £90k from Manufacturing Technology Center
PhD studentship to utilise MTC's state-of-the-art in additive manufacturing for RF frontend systems.

Signal of Opportunity for LEO-based PNT Service with Fast Beam-Steering Antenna Jul 2025 – Aug 2026
Lead with value of £250k from UK Defence and Innovation with GMV UK
- *Commercialisation of Luneburg lens-based ground station antenna developed within the fellowship programme.*

Innovation Launchpad Network+ Research in Residence Aug 2024 – Mar 2026
Lead with value of £50k from UKRI Innovate UK with Satellite Applications Catapult
- *Industrial demonstrator for CubeSat-based beam-steering antenna.*

Small Platform Antennas for CubeSat – UK IC Research Fellowship Jan 2024 – Dec 2025
Lead with value of £275k from Royal Academy of Engineering
- *Novel low-power X-band beam-steering antennas for small satellite platforms and ground stations.*

Grants under review:
- Neuromorphic Integrated Sensing and Communications as Co-lead submitted to UKRI EPSRC.
- Real-time Adaptive Multi-Null Steering Against EM Threats as Lead, stage 2 submitted to UKDI.
- Trusted Integrated Sensing and Communications as Lead, EOI submitted to EPSRC.
- Microparticle Scattering for THz Wavefront Control as Lead, about to be submitted to EPSRC.
- META PRINT SPACE as Co-Lead, about to be submitted to EU Horizon.
- Origami Antennas for CubeSat Platforms as Lead, staged 1 submitted to UKSA.

Research Supervision and Mentoring

Postdoctoral Research Associates:

Dr Pritom Bora	Optically Transparent Metasurfaces	EPSRC funded	2026 – 27
Dr Amir Dayan	Resilient Antennas for Space	ESA funded	2026 – 27
Dr Waleed Ahsan	Secure RIS-based Communications	EPSRC funded	2025 – 26
Dr Amitkumar Patel	Beam-steering Antennas for Ground Station	UKDI and EPSRC funded	2025 – 26

Doctoral Researchers:

Pavan Neelamraju	ML-assisted Intelligent Surfaces	University-funded	2025 – 28
Karthik Gugulothu	Integrated Sensing and Comms for Vehicles	DRIVE CDT funded	2025 – 28
Amitesh Kumar	Optically Transparent Metasurfaces	University-funded	2025 – 28
Damien Beilski	Additive Manufacturing for RF	Industry funded	2025 – 28
Craig Needham (PT)	Resilient Space Infrastructure	Self-funded	2025 – 28

Visiting Academics and Researchers:
- Prof Matthieu Bloch from Georgia Institute of Technology, USA, as part of EPSRC-funded APLUS project.
- Prof Amit Singh from Indian Institute of Technology, Patna, as part of LU IAS visiting fellow programme.

Teaching and Educational Leadership

- Co-developed and co-leading the new Wireless Communications module for third year EE students.
- Co-developed and teaching Robotics Project Management module for second year students throughout school.
- Taught on Advanced Electromagnetics module for second year EE and CE students.
- Supervised 8 UG project students leading to low TRL outputs.
- Delivered guest lectures for different modules and for international institutions.
- Maintained a consistent 4.5 out of 5 across different matrices of teaching quality.

Research Impact, Translation and Commercialisation

- Worked on over 30 consultancy projects since UG studies.
- Work on lens antennas commercialised by Technolog Ltd. with over 15,000 sensors installed across UK.
- Consultancy work for Satraka Ltd on prism-antennas being commercialised for vehicular platforms.
- Prototype for ground-station antenna being finalised for early-stage demonstration.
- Prototype for controlled reception pattern antenna being finalised for early-stage demonstration.
- Active industrial partnership with Lockheed Martin UK, QinetiQ, European Space Agency, GMV UK, Rohde & Schwarz, Satraka Ltd, Technolog, Satellite Applications Catapult, Digital Catapult, and Manufacturing Technology Center.

Academic Leadership, Esteem and Service

- Young Professionals Committee, IEEE ANTEM 2027
- Guest Editor, Special Collection for Nature npj Metamaterials (2026 – present)
- Lead, IEEE Standards Sub-Committee for Metamaterials and Metasurfaces (2025 – present)
- Committee Member, IEEE Standards Committee (2025 – present)
- Co-Lead, Active Metamaterials Special Interest Group, UK Metamaterials Network+ (2024 – present)

- Young Professionals and Technical Programme Committee, IEEE MAPCON 2024 and 2025, India
- Associate Editor for Special Issue in IEEE Open Journal on Antennas and Propagation (2021 – 2022)
- Reviewer for UKRI EPSRC, Royal Academy of Engineering and the Royal Society
- Reviewer for IEEE TAP, AWPL, OJAP, T-MTT, TIM, Scientific Reports, Access, etc.
- Advisor, Atal Innovation Mission, Govt. of India (2017 – 2019)

Selected Awards and Honours

- Sustainable Excellence & Valuable Contribution to School Award, Loughborough University, 2024, 2025 and 2026.
- IEEE Young Professional Best Paper Award, IEEE Microwave, Antennas, and Propagation Conference 2025.
- Young Scientist Award, IEEE Antennas and Propagation Society, 2025.
- Royal Academy of Engineering UK IC Research Fellowship, 2023.
- Doctoral President's Award, Loughborough University, 2022.
- Sir Robert Martin University Prize, Loughborough University, 2020.
- PhD Award for Overall Impact, Loughborough University, 2020.
- Electronics Weekly BrightSparks Award 2020.
- Young Engineer Award, CSIR-CEERI India, 2019.

Selected Journal Articles

Published >80 peer-reviewed papers. Full list available on Google Scholar.

1. J. Lota, P. Neelamraju, A. Demosthenous, A. Bansal, "A Two-Stage RIS Architecture for Decoupled M-FSK Modulation and Beamforming in 28-GHz Bistatic Sensing," IEEE Transactions on Vehicular Technology, 2026 (Accepted).
2. A. Bansal, P. Sharma, J. Chung, "Design and Analysis of Holographic Metasurface Integrated Flexible Antenna for Directional Applications," Transactions on Emerging Telecommunications Technologies, 2026. (Accepted)
3. T. Whittaker, A. Kaye, W. G. Whittow, A. Bansal, "Viscous Jet Printing of RF Devices on Curved Surfaces," Flexible and Printed Electronics (IOP), 2026.
4. M. Lappa, P. Carvill, A. Bansal, et al., "Roadmap on Materials Development and Utilisation in Space," Journal of Physics: Materials, 2026.
5. A. Patel, A. Bansal, et al., "Single and Multi-Beam 360° Pattern Reconfigurable Antenna for 5G Millimeter-wave Communication," IEEE Access, 2026.
6. S. Pope, D. J. Rothe, A. Bansal, "The 2026 Active Metamaterials Roadmap," Journal of Physics D: Applied Physics, 2026.
7. A. Patel, A. Bansal, W. G. Whittow, "Wideband End-Fire Antenna Based on a Modulated Grooved Surface Plasmon Polaritons," Scientific Reports, 2025.
8. L. Poli, A. Bansal, G. Oliveri, et al., "Time-Modulated EM Skins for Integrated Sensing and Communication," IEEE Journal of Selected Topics in Electromagnetics, Antennas and Propagation, 2025.
9. N. Tewari, A. Bansal, et al., "A Novel High Gain Hexagonal Cavity-Backed MIMO SIW Antenna with 3D Homogeneous Lens Loading at Ka-Band," IEEE Antennas and Wireless Propagation Letters, 2025.
10. N. Faisal, V. Rajendran, A. Bansal, et al., "Air Plasma Sprayed Multi-Material Composite Coatings for Enhanced Light Absorption and Thermal Emission," Journal of Thermal Spray Technology, 2025.
11. P. Sharma, A. Bansal, J. Y. Chung, "Conformal Holographic Metasurface-Based Beamforming Antenna using a 3D-Printed Flexible Substrate," Engineering Research Express (IOP), 2025.
12. A. Bansal, C. J. Panagamuwa, W. G. Whittow, "State-of-the-art Millimeter-Wave Beam-Steering Antennas for Beyond 5G and 6G Networks — A Comprehensive Survey," IEEE Antennas and Propagation Magazine, 2024.
13. A. Bansal, W. G. Whittow, "Beam-Scanning Orbital Angular Momentum Beam Circular Leaky Wave Antenna with Half Mode Corrugated Substrate Integrated Waveguide," IEEE Antennas and Wireless Propagation Letters, 2023.
14. A. Bansal, C. J. Panagamuwa, W. G. Whittow, "Fixed Frequency Beam-Steering using Bow-Tie Slot-Based Dielectric Filled Waveguide Antenna Array," Electronics Letters, vol. 59, no. 13, 2023.
15. A. Bansal, C. J. Panagamuwa, W. G. Whittow, "Full 360° Beam Steering Millimeter-Wave Leaky-Wave Antennas Coupled with Bespoke 3D-Printed Dielectric Lenses for 5G Base Stations," Electronics Letters, vol. 59, no. 8, 2023.
16. A. Bansal, C. J. Panagamuwa, W. G. Whittow, "Novel Design Methodology of 3D-Printed Lenses for Travelling Wave Antennas," IEEE Open Journal of Antennas and Propagation, vol. 4, pp. 196–206, 2023.

Key Papers Under Review:

17. W. Ahsan, A. Bansal, "Communication Failure with Reconfigurable Intelligent Surfaces: A Detailed Experimental Campaign," Nature Communication Engineering, 2026.
18. P Bora, A. Kumar, A. Bansal, "Reaction Confining Biogel Lithographing Enabling High Fidelity Pattering and In Situ Metal Capture in Transparent Conductors," Science, 2026.
19. P. Aggarwal, N. Tewari, S. Srivastava, A. Bansal, "Compact Ultra-Wideband High Gain Cavity-backed SIW Metasurface Antenna," IEEE Antennas and Wireless Propagation Letters, 2026.
20. T. Whittaker, A. Bansal, W. G. Whittow, "Anisotropic and Diamagnetic Material Characterisation with a Low-Cost 3D Printed Open Resonator Using Curved FSS," IEEE Transactions on Instrumentation and Measurement, 2026.
21. E. Vassos, T. Whittaker, A. Jabbar, W. G. Whittow, A. Bansal, "Fully 3D Printed Wideband Metasurface Folded Reflectarray Antenna," IEEE Transactions on Antennas and Propagation, 2026.
22. S. Ghosh, A. Bansal, C. Saha, "Comprehensive Analytical Characterization of Statistically Distributed Slotted SIW-based Leaky Wave Antenna for SLL and HPBW Reduction," IEEE Transactions on Antennas and Propagation, 2026.

References Available on Request